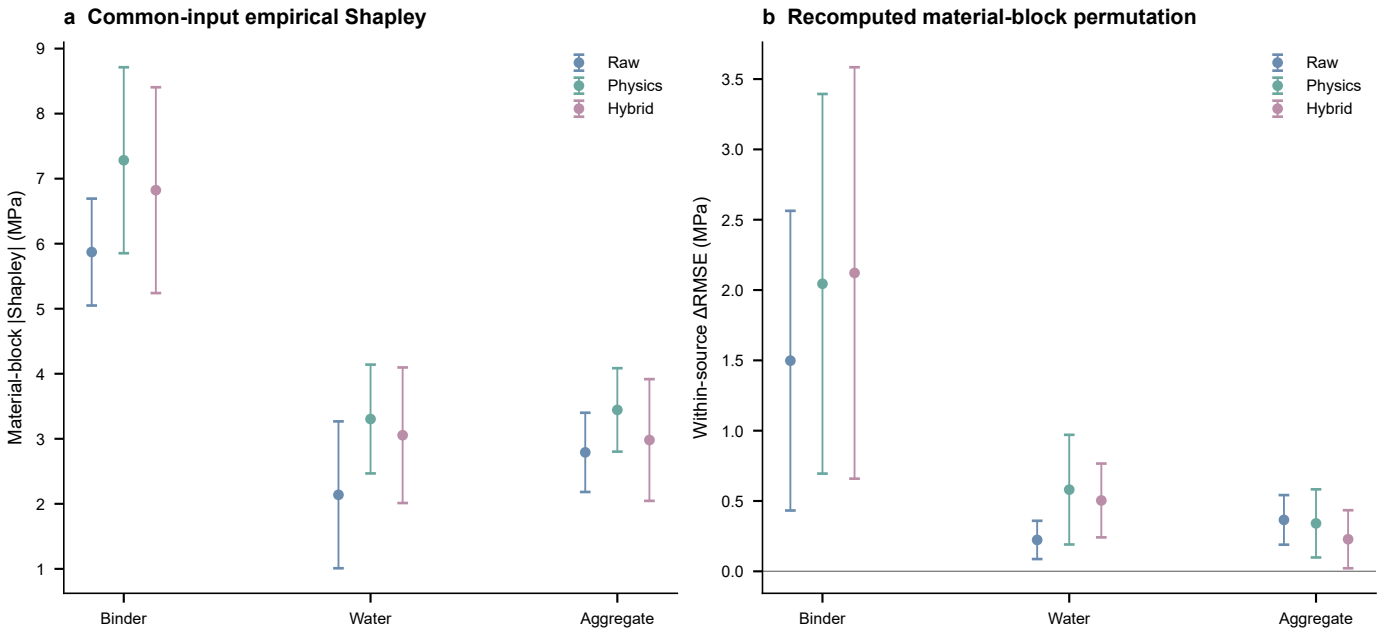

a Common-input empirical Shapley
Material-block |Shapley| (MPa)
9
8
7
6
5
4
3
2
1
Raw
Physics
Hybrid
Binder
Water
Aggregate
b Recomputed material-block permutation
Within-source ΔRMSE (MPa)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Raw
Physics
Hybrid
Binder
Water
Aggregate